Frequent Subgraph Mining of Functional Interactions Patterns Across Multiple Cancers

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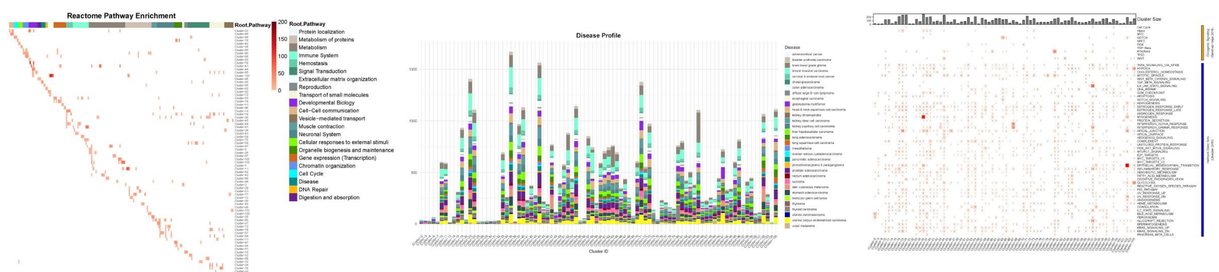
Abstract

Motivation: Molecular mechanisms characterizing cancer development and progression are complex and process through thousands of interacting elements in the cell. Understanding the underlying structure of interactions requires the integration of cellular networks with extensive combinations of dysregulation patterns. Recent pan-cancer studies focused on identifying common dysregulation patterns in a confined set of pathways or targeting a manually curated set of genes. However, the complex nature of the disease presents a challenge for finding pathways that would constitute a basis for tumor progression and requires evaluation of subnetworks with functional interactions. Uncovering these relationships is critical for translational medicine and the identification of future therapeutics.

Materials & Methods: We describe an integrative -omics approach to pan-cancer analysis using Frequent Subgraph Mining (FSM) coupled with biased random walks utilizing genomic alterations, gene expression, and PPI networks. We use FSM to identify frequently occurring interaction patterns to provide a better understanding of functional alterations across multiple cancer types while accounting for the complex interaction topology of cancer. Our goal is to integrate PPI networks with somatic alterations and gene expression profiles to infer molecular networks representing dysregulation in cancers. More specifically, we extract subnetworks that are frequent in the population and in close proximity to the mutated genes. We investigate TCGA samples for 32 cancer types. We present patient clusters across all cancer types as well as patient classifications of individual cancers based on these networks. We identify mechanisms that are shared across tumor types and unique to individual cancers.

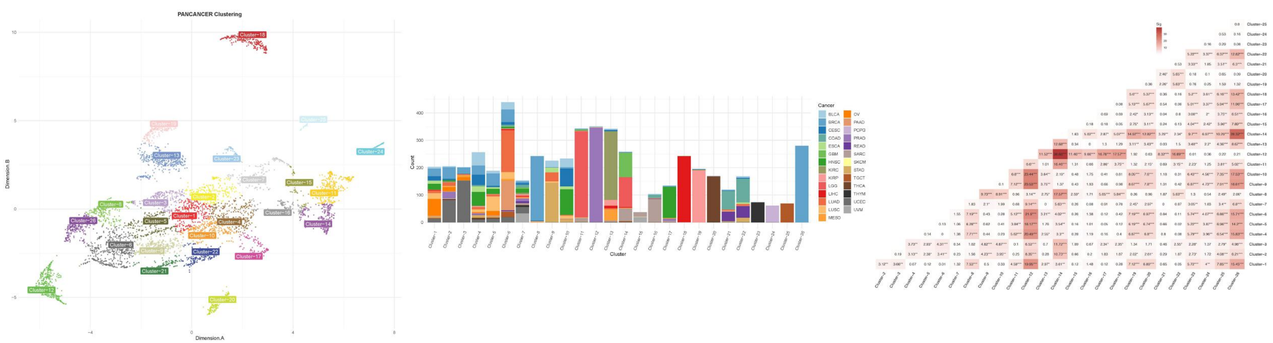
Results: Investigating pathway enrichment results of identified frequent subgraphs (fsg) using Reactome recapitulated previously defined cancer associated pathways such as *PI3K* signaling, Cytokine Signaling, Extracellular Matrix Organization in addition to previously not associated pathways. Furthermore disease overrepresentation of mined pathways showed highly heterogeneous topology of frequently interacting genes **(Fig.1).** Clustering of samples based on mined fsg showed separation correlated highly with tissue type hence significant survival differences are observed **(Fig.2).** Furthermore, we have used the mined fsg in a disease specific mode to investigate whether the mined fsgs can capture local topologies. We were able to extract clusters significantly different in terms of survival in 2 disease types; LGG and Uterine cancers which suggest that the mined subgraphs can indeed capture patterns specific to cancer types **(Fig.3).** However the utility of using fsgs mined across PANCANCER dataset in single cancer types should be further evaluated

FSG Profiles



(a) Reactome Enrichment **(b)** Disease Overrepresentation **(c)** Gene Set Overlap

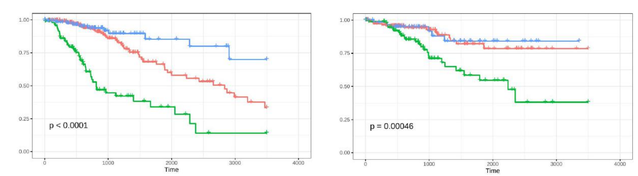
Figure 1: Enrichment and overrepresentation profiles of identified frequent subgraphs showing the heterogeneity of mined interaction patterns.



(a) Clustering of samples **(b)** Disease Profiles of Clusters **(c)** Survival Comparison

Figure 2: Clustering of TCGA samples based on FSG profiles using density based clustering on low dimensional embedding showing significant survival differences and disease specific profiles

Individual Types



(a) Lower Grade Glioma **(b)** Uterine

Figure 3: Survival comparison of identified cluster using frequent subgraphs in a cancer specific fashion

Conclusion

We have investigated the utility of frequent subgraph mining approaches in cancer subtype discovery where we showed that complex topology of protein-protein interactions can be extracted relatively efficiently. However, it is non-trivial to assess the significance of the mined subgraphs since relatively large number of frequent patterns are extracted hence inherent noise is high. Nevertheless, overlap of mined subgraphs recapitulated and extended previously defined cancer associated pathways and modules. Furthermore dysregulation quantification of mined subgraphs revealed disease associated patterns where multiple disease subgroups are clustered together with significant survival differences.

References

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